STAT GU4204 — Statistical Inference

Final Exam — Practice Version A: **Solutions**

Spring 2026

Solutions show full derivations. Final answers are boxed. Notation matches the exam.

Part A — Short Conceptual

Problem 1. Cramér–Rao Lower Bound **(3 points)**

If $T(X_1, \ldots, X_n)$ is an unbiased estimator of $g(\theta)$ (so $E_\theta[T] = g(\theta)$ for all θ), then under the regularity conditions

$$\mathrm{Var}_\theta(T) \;\geq\; \frac{[g'(\theta)]^2}{n\,I(\theta)},$$

where $I(\theta) = E_\theta\Big[\big(\frac{\partial}{\partial\theta}\log f(X;\theta)\big)^2\Big]$ is the Fisher information in a single observation. In the special case $g(\theta) = \theta$, the bound becomes $\mathrm{Var}_\theta(T) \geq 1/[nI(\theta)]$.

One regularity condition. Any one of the following:

- (A.1) The support $\{x : f(x;\theta) > 0\}$ does not depend on θ.
- (A.2) Differentiation under the integral sign is valid: $\frac{\partial}{\partial\theta}\int W(x)f(x;\theta)\,dx = \int W(x)\frac{\partial}{\partial\theta}f(x;\theta)\,dx$.
- (A.3) $\frac{\partial}{\partial\theta}\log f(x;\theta)$ exists and is finite for every x in the support.

Answer. $\mathrm{Var}_\theta(T) \geq [g'(\theta)]^2/[nI(\theta)]$, with one regularity condition such as "support of f does not depend on θ."

Problem 2. Neyman–Pearson Lemma **(3 points)**

Setup. Simple null vs. simple alternative: $H_0 : \theta = \theta_0$ vs. $H_1 : \theta = \theta_1$, with joint densities f_0 and f_1.

Statement. For any constant $k \geq 0$, the test δ^* defined by

- Reject H_0 if $f_1(\mathbf{x}) > k\,f_0(\mathbf{x})$,
- Do not reject if $f_1(\mathbf{x}) < k\,f_0(\mathbf{x})$,
- Either decision is allowed when $f_1(\mathbf{x}) = k\,f_0(\mathbf{x})$,

is most powerful at its level: for any other test δ with $\alpha(\delta) \leq \alpha(\delta^*)$, we have $\beta(\delta) \geq \beta(\delta^*)$ (equivalently, $\pi(\theta_1;\delta) \leq \pi(\theta_1;\delta^*)$).

Answer. The likelihood-ratio test that rejects when $f_1/f_0 > k$ minimizes Type II error among all level-$\alpha(\delta^*)$ tests.

Problem 3. Sufficient Statistics **(2 points)**

A statistic $T = \varphi(X_1, \ldots, X_n)$ is *sufficient* for θ if the conditional distribution of $(X_1, \ldots, X_n)$ given $T = t$ does not depend on θ.

Factorization criterion. T is sufficient for θ if and only if the joint density/pmf factors as

$$f_n(\mathbf{x}; \theta) = u(\mathbf{x}) \cdot v(\varphi(\mathbf{x}), \theta)$$

for some non-negative functions u (free of θ) and v (depending on the data only through T).

Answer. Sufficiency means the conditional distribution of the data given T is free of θ; factorization: $f_n(\mathbf{x}; \theta) = u(\mathbf{x})v(\varphi(\mathbf{x}), \theta)$.

Problem 4. Test/CI Duality **(2 points)**

For each $\theta_0 \in \Omega$, let $\mathcal{A}(\theta_0)$ be the acceptance region of a level-α_0 test of $H_0 : \theta = \theta_0$. Define the random set

$$\mathcal{S}(\mathbf{X}) = \{\theta_0 : \mathbf{X} \in \mathcal{A}(\theta_0)\}.$$

Then $\mathcal{S}(\mathbf{X})$ is a $(1 - \alpha_0)$ confidence set for θ, because $P_\theta(\theta \in \mathcal{S}(\mathbf{X})) = P_\theta(\mathbf{X} \in \mathcal{A}(\theta)) \geq 1 - \alpha_0$. Conversely, given a confidence set $\mathcal{S}$, the test "reject $H_0 : \theta = \theta_0$ iff $\theta_0 \notin \mathcal{S}(\mathbf{X})$" has level α_0.

Answer. The $(1 - \alpha_0)$ CI is the set of null values θ_0 that the level-α_0 test would NOT reject; equivalently, invert the acceptance region.

Part B — Mid-Length Problems

Problem 5. Exponential MLE, Delta Method, CI **(15 points)**

(a) *(5 pts)* The likelihood and log-likelihood are

$$L_n(\theta) = \prod_{i=1}^{n} \theta e^{-\theta X_i} = \theta^n \exp\Big(- \theta \textstyle\sum_i X_i \Big), \qquad \ell_n(\theta) = n \log \theta - \theta \textstyle\sum_{i=1}^n X_i.$$

Score: $\dot{\ell}_n(\theta) = n/\theta - \sum X_i$. Setting equal to zero gives $\theta = n/\sum X_i = 1/\bar{X}_n$. The second derivative is $-n/\theta^2 < 0$, so this is the unique maximum.

Answer. $\widehat{\theta}_n = 1/\bar{X}_n$.

(b) *(3 pts)* By the Weak Law of Large Numbers, $\bar{X}_n \xrightarrow{P} E_\theta[X_1] = 1/\theta$. The map $g(t) = 1/t$ is continuous on $(0, \infty)$, so by the continuous mapping theorem,

$$\widehat{\theta}_n = 1/\bar{X}_n \xrightarrow{P} 1/(1/\theta) = \theta.$$

Answer. $\widehat{\theta}_n \xrightarrow{P} \theta$ by WLLN + continuous mapping theorem; hence $\widehat{\theta}_n$ is consistent.

(c) *(4 pts)* By the Central Limit Theorem applied to $\bar{X}_n$ (with mean $1/\theta$ and variance $1/\theta^2$):

$$\sqrt{n}\big(\bar{X}_n - 1/\theta\big) \xrightarrow{d} N(0, 1/\theta^2).$$

Apply the delta method with $g(t) = 1/t$, so $g'(t) = -1/t^2$ and $g'(1/\theta) = -\theta^2$. Thus

$$\sqrt{n}\big(\widehat{\theta}_n - \theta\big) = \sqrt{n}\big(g(\bar{X}_n) - g(1/\theta)\big) \xrightarrow{d} N\big(0,\ [g'(1/\theta)]^2 \cdot 1/\theta^2\big) = N(0,\ \theta^4 \cdot 1/\theta^2) = N(0, \theta^2).$$

Answer. $\sqrt{n}(\widehat{\theta}_n - \theta) \xrightarrow{d} N(0, \theta^2)$, i.e. $\tau^2(\theta) = \theta^2$.

(d) *(3 pts)* From (c), for large n, $\widehat{\theta}_n \approx N(\theta, \theta^2/n)$. Replacing θ by the consistent estimator $\widehat{\theta}_n$ in the asymptotic standard error is justified by Slutsky's theorem (since $\widehat{\theta}_n/\theta \xrightarrow{P} 1$). An approximate 95% CI for θ is therefore

$$\widehat{\theta}_n \pm z_{0.025} \cdot \frac{\widehat{\theta}_n}{\sqrt{n}} = \widehat{\theta}_n \left(1 \pm \frac{1.96}{\sqrt{n}}\right).$$

Answer. $\Big[\widehat{\theta}_n\big(1 - 1.96/\sqrt{n}\big),\ \widehat{\theta}_n\big(1 + 1.96/\sqrt{n}\big)\Big]$, where $\widehat{\theta}_n = 1/\bar{X}_n$.

Problem 6. Bernoulli with Beta Prior; Sufficiency **(15 points)**

Let $T = \sum_{i=1}^n X_i$, so $T \mid p \sim \text{Binomial}(n, p)$ and the likelihood is $L(p) \propto p^t(1-p)^{n-t}$.

(a) *(5 pts)* The Beta(α, β) prior has density $\pi(p) \propto p^{\alpha-1}(1-p)^{\beta-1}$ on $(0, 1)$. By Bayes' rule,

$$\pi(p \mid \mathbf{X}) \ \propto\ L(p)\,\pi(p) \ \propto\ p^{\alpha+t-1}(1-p)^{\beta+n-t-1}.$$

This is the kernel of a Beta distribution.

Answer. $p \mid \mathbf{X} \ \sim\ \text{Beta}\big(\alpha + T,\ \beta + n - T\big)$, where $T = \sum_{i=1}^n X_i$.

(b) *(3 pts)* Under squared-error loss the Bayes estimator is the posterior mean. For a Beta(a, b) distribution the mean is $a/(a+b)$, so

$$\widehat{p}_{\text{Bayes}} = \frac{\alpha + T}{\alpha + \beta + n}.$$

Answer. $\widehat{p}_{\text{Bayes}} = (\alpha + \sum X_i)/(\alpha + \beta + n)$.

(c) *(4 pts)* The joint pmf of $(X_1, \ldots, X_n)$ is

$$f_n(\mathbf{x} \mid p) = \prod_{i=1}^n p^{x_i}(1-p)^{1-x_i} = p^t(1-p)^{n-t}, \qquad t = \sum x_i,$$

where each $x_i \in \{0,1\}$. Take

$$u(\mathbf{x}) = 1 \cdot \mathbf{1}\{x_i \in \{0,1\} \text{ for all } i\}, \qquad v(t,p) = p^t(1-p)^{n-t}.$$

Then $f_n(\mathbf{x} \mid p) = u(\mathbf{x}) \cdot v(T(\mathbf{x}), p)$, which is the factorization criterion.

Answer. $T = \sum X_i$ is sufficient for p by factorization with $u \equiv 1$ and $v(t,p) = p^t(1-p)^{n-t}$.

(d) *(3 pts)* Rewrite the Bayes estimator:

$$\widehat{p}_{\text{Bayes}} = \frac{\alpha + n\bar{X}_n}{\alpha+\beta+n} = \frac{n}{\alpha+\beta+n}\bar{X}_n + \frac{\alpha}{\alpha+\beta+n}.$$

As $n \to \infty$, the first coefficient tends to 1, the second to 0, and $\bar{X}_n \xrightarrow{P} p$ by the WLLN. Hence (using continuous mapping / Slutsky):

$$\widehat{p}_{\text{Bayes}} \xrightarrow{P} p.$$

The data dominates the prior asymptotically; equivalently, the Bayes estimator converges to the MLE $\bar{X}_n$.

Answer. $\widehat{p}_{\text{Bayes}} \xrightarrow{P} p$ as $n \to \infty$ (the prior washes out; estimator $\to$ MLE).

Part C — Major Problems

Problem 7. Two-Sample t-Test and F-Test **(20 points)**

Throughout, $m = n = 8$, $\bar{X} = 23.5$, $\bar{Y} = 21.0$, $S_X^2 = 21$, $S_Y^2 = 14$.

(a) *(10 pts)*

(i) Test statistic. Under the common-variance model, the two-sample t-statistic is

$$U_{m,n} = \frac{\sqrt{m+n-2}\,(\bar{X}-\bar{Y})}{\sqrt{\left(\frac{1}{m}+\frac{1}{n}\right)(S_X^2+S_Y^2)}}.$$

Under $H_0 : \mu_X = \mu_Y$, $U_{m,n} \sim t_{m+n-2} = t_{14}$.

(ii) Plug in. Numerator: $\sqrt{14}\,(23.5 - 21.0) = 2.5\sqrt{14}$. Denominator: $\sqrt{(\frac{1}{8}+\frac{1}{8})(21+14)} = \sqrt{\frac{1}{4}\cdot 35} = \sqrt{35/4} = \frac{1}{2}\sqrt{35}$. Therefore

$$U_{m,n} = \frac{2.5\sqrt{14}}{\frac{1}{2}\sqrt{35}} = 5\sqrt{14/35} = 5\sqrt{2/5} = \sqrt{25\cdot 2/5} = \sqrt{10} \approx 3.162.$$

(iii) Decision. For a two-sided level-0.05 test, reject if $|U_{m,n}| > t_{0.025,14} = 2.145$. Since $|\sqrt{10}| \approx 3.162 > 2.145$, we reject H_0.

Answer. $U_{m,n} = \sqrt{10} \approx 3.162$. Critical value $t_{0.025,14} = 2.145$. **Reject** H_0 at $\alpha_0 = 0.05$: there is significant evidence that $\mu_X \neq \mu_Y$.

(b) *(5 pts)* The pooled variance estimator is

$$\widehat{\sigma}^2 = \frac{S_X^2 + S_Y^2}{m+n-2} = \frac{35}{14} = 2.5.$$

The standard error of $\bar{X} - \bar{Y}$ is

$$\mathrm{SE}(\bar{X} - \bar{Y}) = \sqrt{\widehat{\sigma}^2\big(\tfrac{1}{m} + \tfrac{1}{n}\big)} = \sqrt{2.5 \cdot \tfrac{1}{4}} = \sqrt{0.625} = \tfrac{\sqrt{10}}{4}.$$

A 95% CI for $\mu_X - \mu_Y$ is therefore

$$(\bar{X} - \bar{Y}) \pm t_{0.025,14} \cdot \mathrm{SE} = 2.5 \pm 2.145 \cdot \tfrac{\sqrt{10}}{4}.$$

Numerically, $2.145 \cdot \sqrt{10}/4 \approx 2.145 \cdot 0.7906 \approx 1.696$.

Answer. 95% CI for $\mu_X - \mu_Y$: $\big[2.5 - 2.145 \cdot \frac{\sqrt{10}}{4},\ 2.5 + 2.145 \cdot \frac{\sqrt{10}}{4}\big] \approx [0.804,\ 4.196]$. Note 0 is not in this interval, consistent with rejecting H_0 in part (a).

(c) *(5 pts)* The F-statistic for comparing variances is

$$V_{m,n} = \frac{S_X^2/(m-1)}{S_Y^2/(n-1)} = \frac{21/7}{14/7} = \frac{3}{2} = 1.5.$$

Under $H_0 : \sigma_X^2 = \sigma_Y^2$, $V_{m,n} \sim F_{m-1,n-1} = F_{7,7}$.

For a two-sided level-$\alpha_0 = 0.10$ test, reject if

$$V \leq F_{0.95,7,7} = 1/F_{0.05,7,7} = 1/3.79 \approx 0.264 \quad \text{or} \quad V \geq F_{0.05,7,7} = 3.79.$$

Since $0.264 < 1.5 < 3.79$, we fail to reject H_0.

Answer. $V = 1.5$, rejection region $V \leq 0.264$ or $V \geq 3.79$. **Fail to reject** H_0 at $\alpha_0 = 0.10$: no significant evidence that $\sigma_X^2 \neq \sigma_Y^2$. The common-variance assumption used in (a)–(b) is not contradicted by the data.

Problem 8. Neyman–Pearson, Power, Sample Size **(20 points)**

(a) *(8 pts)* The joint density under θ is

$$f_\theta(\mathbf{x}) = (2\pi)^{-n/2} \exp\Big(-\tfrac{1}{2} \textstyle\sum_i (x_i - \theta)^2 \Big).$$

The likelihood ratio is

$$\frac{f_1(\mathbf{x})}{f_0(\mathbf{x})} = \exp\Big(-\tfrac{1}{2} \textstyle\sum_i [(x_i - 1)^2 - x_i^2] \Big) = \exp\Big(\textstyle\sum_i x_i - \tfrac{n}{2} \Big) = \exp\big(n\bar{x} - \tfrac{n}{2} \big).$$

By the Neyman–Pearson Lemma, the most powerful level-α_0 test rejects H_0 when $f_1/f_0 > k$ for some $k > 0$. Since $\exp(n\bar{x} - n/2)$ is strictly increasing in $\bar{x}$, this is equivalent to rejecting when

$$\bar{X}_n > c \qquad \text{for some constant } c.$$

The constant c is chosen so that $P_{\theta=0}(\bar{X}_n > c) = \alpha_0$.

Answer. The MP test rejects when $\bar{X}_n > c$, with c chosen to make the size $= \alpha_0$.

(b) *(4 pts)* Under H_0 ($\theta = 0$), $\bar{X}_n \sim N(0, 1/n)$, so $\sqrt{n}\,\bar{X}_n \sim N(0,1)$. We need

$$P_0(\bar{X}_n > c) = P\big(Z > \sqrt{n}\,c\big) = 0.05 \implies \sqrt{n}\,c = z_{0.05} = 1.645.$$

With $n = 16$, $\sqrt{n} = 4$, so $c = 1.645/4$.

Answer. $c = 1.645/4 \approx 0.4113$.

(c) *(5 pts)* Under $\theta = 1$, $\bar{X}_n \sim N(1, 1/n)$, so $\sqrt{n}(\bar{X}_n - 1) \sim N(0,1)$. The power is

$$\pi(1) = P_{\theta=1}\big(\bar{X}_n > c\big) = P\big(Z > \sqrt{n}(c-1)\big) = 1 - \Phi\big(\sqrt{n}\,c - \sqrt{n}\big) = \Phi\big(\sqrt{n} - \sqrt{n}\,c\big).$$

Since $\sqrt{n}\,c = z_{0.05} = 1.645$ and $\sqrt{n} = 4$,

$$\pi(1) = \Phi(4 - 1.645) = \Phi(2.355).$$

The Type II error probability is $\beta = 1 - \pi(1) = 1 - \Phi(2.355)$.

Numerically $\Phi(2.355) \approx 0.9907$ (between $\Phi(2.326) = 0.99$ and $\Phi(2.576) = 0.995$), so $\beta \approx 0.0093$.

Answer. $\pi(1) = \Phi(4 - 1.645) = \Phi(2.355) \approx 0.9907$. $\quad \beta = 1 - \pi(1) \approx 0.0093$.

(d) *(3 pts)* For the analogous test with sample size n, the rejection threshold is $c_n = z_{0.05}/\sqrt{n}$ and the power is

$$\pi(1; n) = \Phi\big(\sqrt{n} - z_{0.05}\big) = \Phi(\sqrt{n} - 1.645).$$

We need $\pi(1; n) \geq 0.99 = \Phi(z_{0.01})$, i.e. $\sqrt{n} - 1.645 \geq z_{0.01} = 2.326$, hence

$$\sqrt{n} \geq 1.645 + 2.326 = 3.971 \quad \iff \quad n \geq 3.971^2 \approx 15.77.$$

The smallest integer n satisfying this is $n^* = 16$. (The current sample size already meets the 0.99 power requirement, consistent with $\pi(1) \approx 0.9907$ from part (c).)

Answer. $n^* = 16$.

Problem 9. Poisson MLE, Fisher Information, LRT **(20 points)**

(a) *(5 pts)* The log-likelihood (dropping terms free of λ) is

$$\ell_n(\lambda) = -n\lambda + \log(\lambda) \sum_{i=1}^{n} X_i + \text{const}.$$

Differentiating: $\dot{\ell}_n(\lambda) = -n + (\sum X_i)/\lambda$. Setting $= 0$ gives $\lambda = \bar{X}_n$. The second derivative $-(\sum X_i)/\lambda^2 < 0$, confirming a maximum.

Unbiasedness: $E_\lambda[\widehat{\lambda}_n] = E_\lambda[\bar{X}_n] = \frac{1}{n}\sum_{i=1}^n E_\lambda[X_i] = \frac{1}{n}(n\lambda) = \lambda$.

Answer. $\widehat{\lambda}_n = \bar{X}_n$. It is unbiased: $E_\lambda[\widehat{\lambda}_n] = \lambda$.

(b) *(5 pts)* For a single observation $X \sim \text{Poisson}(\lambda)$: $\log f(x;\lambda) = -\lambda + x\log\lambda - \log(x!)$, so the score is $\dot{\ell}(x;\lambda) = -1 + x/\lambda$. The Fisher information is

$$I(\lambda) = E_\lambda\Big[\dot{\ell}(X;\lambda)^2\Big] = \text{Var}_\lambda(\dot{\ell}) = \text{Var}_\lambda(X/\lambda) = \frac{\text{Var}_\lambda(X)}{\lambda^2} = \frac{\lambda}{\lambda^2} = \frac{1}{\lambda}.$$

For the full sample (i.i.d.): $I_n(\lambda) = nI(\lambda) = n/\lambda$.

Efficiency check. The CRLB for unbiased estimators of λ is $1/[nI(\lambda)] = \lambda/n$. The variance of $\widehat{\lambda}_n$ is

$$\text{Var}_\lambda(\widehat{\lambda}_n) = \text{Var}_\lambda(\bar{X}_n) = \frac{\text{Var}_\lambda(X_1)}{n} = \frac{\lambda}{n}.$$

This equals the CRLB, so $\widehat{\lambda}_n$ achieves the bound and is therefore efficient.

Answer. $I(\lambda) = 1/\lambda$, $I_n(\lambda) = n/\lambda$, CRLB $= \lambda/n = \text{Var}_\lambda(\widehat{\lambda}_n)$. Yes, $\widehat{\lambda}_n$ is efficient.

(c) *(6 pts)* The likelihood is (constant $C(\mathbf{x}) = 1/\prod x_i!$)

$$L_n(\lambda;\mathbf{x}) = e^{-n\lambda}\lambda^{n\bar{x}} \cdot C(\mathbf{x}).$$

Under H_0 the supremum is $L_n(\lambda_0)$. Over all of Ω the supremum is $L_n(\widehat{\lambda}_n) = L_n(\bar{x})$. Therefore

$$\Lambda(\mathbf{X}) = \frac{L_n(\lambda_0)}{L_n(\widehat{\lambda}_n)} = \frac{e^{-n\lambda_0}\lambda_0^{n\widehat{\lambda}_n}}{e^{-n\widehat{\lambda}_n}\widehat{\lambda}_n^{n\widehat{\lambda}_n}} = \exp\Big(-n(\lambda_0 - \widehat{\lambda}_n)\Big)\left(\frac{\lambda_0}{\widehat{\lambda}_n}\right)^{n\widehat{\lambda}_n}.$$

Take logs:

$$\log\Lambda = -n(\lambda_0 - \widehat{\lambda}_n) + n\widehat{\lambda}_n\log(\lambda_0/\widehat{\lambda}_n) = n(\widehat{\lambda}_n - \lambda_0) - n\widehat{\lambda}_n\log(\widehat{\lambda}_n/\lambda_0).$$

Therefore

$$-2\log\Lambda(\mathbf{X}) = 2n\left[\widehat{\lambda}_n\log\frac{\widehat{\lambda}_n}{\lambda_0} - \big(\widehat{\lambda}_n - \lambda_0\big)\right].$$

Answer. $\Lambda(\mathbf{X}) = e^{-n(\lambda_0-\widehat{\lambda}_n)}(\lambda_0/\widehat{\lambda}_n)^{n\widehat{\lambda}_n}$. Equivalently, $-2\log\Lambda(\mathbf{X}) = 2n\big[\widehat{\lambda}_n\log(\widehat{\lambda}_n/\lambda_0) - (\widehat{\lambda}_n - \lambda_0)\big]$.

(d) *(4 pts)* Wilks' theorem: under H_0 with H_0 specifying $k = 1$ coordinate, as $n \to \infty$,

$$-2\log\Lambda(\mathbf{X}) \xrightarrow{d} \chi^2_1.$$

A level-$\alpha_0 = 0.05$ asymptotic test therefore rejects H_0 when

$$-2\log\Lambda(\mathbf{X}) \;>\; \chi^2_{0.05,1} = 3.841.$$

Answer. Reject H_0 at asymptotic level 0.05 when $-2\log\Lambda(\mathbf{X}) > 3.841$. (Limiting distribution: χ^2_1.)

End of Solutions — Total Points: $10 + 30 + 60 = 100$.